

Harmonic Structure in Superconductor Energy Gaps and Black Hole Quasinormal Modes: An Empirical Observation

Jaie Parker

Independent Researcher, Australia

Correspondence: [email to be added]

Abstract

We report a previously unnoticed empirical pattern: when superconductor energy gap ratios ($2\Delta/k_B T_c$) are octave-reduced — repeatedly divided by 2 until they fall within the interval $[1, 2)$ — the resulting values cluster on simple integer (consonance) ratios with a precision that is statistically incompatible with chance. Among 10 published gap ratios spanning conventional BCS and strong-coupling superconductors, 3 values coincide exactly with consonance ratios (0.00% deviation), and 6 of 10 fall within 1% of a consonance point, with all 10 within 5%. The same pattern appears independently in black hole quasinormal mode frequency ratios and in fundamental physical constants. A Monte Carlo simulation ($N = 100,000$) comparing 24 observed values against uniform random draws from the same range yields a combined p-value of 6×10^{-5} (Cohen's $d = 3.30$). A binomial test on the fraction of values within 1% of consonance gives $p = 7.9 \times 10^{-8}$. No theoretical framework is assumed or proposed; we present the observation, the statistics, and falsifiable predictions for future measurement.

Keywords: superconductor energy gap, BCS theory, quasinormal modes, consonance, frequency ratios, statistical anomaly

1. Introduction

1.1 Superconductor Energy Gaps

The Bardeen-Cooper-Schrieffer (BCS) theory of superconductivity [1] predicts a universal ratio between the zero-temperature energy gap $2\Delta(0)$ and the critical temperature T_c :

$$2\Delta(0) / k_B T_c = 3.528$$

This ratio is one of the celebrated quantitative successes of BCS theory. Real materials deviate from this prediction depending on coupling strength: weak-coupling superconductors (Sn, Al, In) cluster near the BCS value, while strong-coupling materials (Pb, Hg, Nb₃Sn) exhibit ratios ranging from 3.8 to 4.6 [2,3]. These deviations are well understood within Eliashberg theory [4] and reflect the detailed structure of the electron-phonon interaction.

1.2 Black Hole Quasinormal Modes

Black holes, when perturbed, ring down through characteristic quasinormal modes (QNMs) whose complex frequencies are determined entirely by the black hole's mass, charge, and angular momentum [5]. For Schwarzschild black holes, the QNM frequencies increase monotonically with the angular momentum number l . The ratios between successive QNM frequencies are dimensionless numbers of order unity that encode fundamental properties of the black hole spacetime geometry [6].

1.3 Octave Reduction

Octave reduction is a standard operation in music theory, signal processing, and psychoacoustics. Given a frequency ratio $r > 1$, the octave-reduced value is:

$$r_{\text{oct}} = r / 2^n, \text{ where } n = \text{floor}(\log_2(r))$$

such that r_{oct} is in $[1, 2)$. This operation preserves the pitch class of a frequency ratio — its position within one octave — while discarding information about which octave it occupies. It is the frequency-domain analogue of taking a value modulo 1 in log-frequency space.

1.4 The Observation

We observe that when superconductor gap ratios are octave-reduced, they fall on or very near the simple integer ratios that define musical consonance. The same pattern appears in black hole QNM frequency ratios and in ratios formed from fundamental physical constants. This paper presents the data, quantifies the statistical significance, and offers predictions.

2. Method

2.1 Data Sources

Superconductor gap ratios were taken from standard references in condensed matter physics [2,3,7,8]. All values are published experimental measurements or, in the case of the BCS theoretical value, the analytic weak-coupling limit. Black hole QNM frequencies were taken from Leaver [9], Berti et al. [6], and the LIGO GW150914 analysis [10]. Fundamental constant ratios use CODATA 2018 recommended values [11] and the hydrogen 21 cm line frequency.

No selection was performed on the data: we included all conventional elemental superconductors with well-established gap ratios (Sn, Al, In, Ta, Nb, Pb, Hg), the BCS theoretical value, and the two most-studied non-elemental superconductors with reliable gap measurements (Nb₃Sn, MgB₂). For QNMs, we included the Schwarzschild and maximal Kerr cases as the two extremes of the parameter space, plus the first gravitational wave observation.

2.2 Consonance Set

We define the consonance set C as the 13 frequency ratios within one octave that correspond to intervals in standard just intonation:

Ratio	Decimal	Musical Interval
1:1	1.000	Unison
16:15	1.067	Minor second
9:8	1.125	Major second
6:5	1.200	Minor third
5:4	1.250	Major third
4:3	1.333	Perfect fourth
7:5	1.400	Tritone
3:2	1.500	Perfect fifth
8:5	1.600	Minor sixth
5:3	1.667	Major sixth
7:4	1.750	Harmonic seventh
9:5	1.800	Minor seventh
15:8	1.875	Major seventh
2:1	2.000	Octave

This set is not arbitrary. These ratios are the complete set of superparticular and simple-integer ratios with denominator ≤ 8 within one octave, and they correspond precisely to the intervals that exhibit frequency-locking in coupled oscillator systems [12,13]. The number of consonance points (14, including both endpoints) and the typical spacing between them (~ 0.05 - 0.10 in $[1,2]$) establish the baseline

expectation: a random value in $[1,2]$ has approximately a 26% chance of falling within 1% of at least one consonance point, and approximately a 5.2% chance of falling within 0.2%.

2.3 Proximity Metric

For an observed octave-reduced ratio r_{oct} , the deviation from the nearest consonance ratio c^* is:

$$\text{delta}(\%) = |r_{\text{oct}} - c^*| / c^* \times 100$$

where $c^* = \text{argmin}_{\{c \in C\}} |r_{\text{oct}} - c|$.

2.4 Monte Carlo Simulation

To assess significance, we performed a Monte Carlo simulation with $N = 100,000$ trials. In each trial, we drew the same number of observations as in our dataset ($n = 24$ for the combined set) from a uniform distribution on $[1, 2)$ and computed the mean proximity to the nearest consonance point. The p-value is the fraction of trials in which the simulated mean proximity was less than or equal to the observed mean proximity.

2.5 Binomial Test

As a complementary test, we counted the number of observed values falling within 1% of a consonance point and compared this to the binomial expectation under the null hypothesis of uniform distribution. The probability of any single random value in $[1, 2)$ falling within 1% of at least one of the 14 consonance points is $p_0 = 0.26$, calculated from the total measure of the union of 1%-neighborhoods around each consonance point within $[1, 2)$.

3. Results: Superconductor Gap Ratios

3.1 Data Table

Material	$2\Delta/k_B T_c$	Octave-Reduced	Nearest Consonance	Interval Name	Deviation (%)
BCS Theory	3.528	1.764	7:4 = 1.750	Harmonic 7th	0.80
Tin (Sn)	3.500	1.750	7:4 = 1.750	Harmonic 7th	0.00
Aluminum (Al)	3.530	1.765	7:4 = 1.750	Harmonic 7th	0.86

Material	$2\Delta/k_B T_c$	Octave-Reduced	Nearest Consonance	Interval Name	Deviation (%)
Niobium (Nb)	3.800	1.900	15:8 = 1.875	Major 7th	1.33
Lead (Pb)	4.300	1.075	9:8 = 1.125	Major 2nd	4.44
Mercury (Hg)	4.600	1.150	9:8 = 1.125	Major 2nd	2.22
Nb3Sn	4.400	1.100	9:8 = 1.125	Major 2nd	2.22
Indium (In)	3.600	1.800	9:5 = 1.800	Minor 7th	0.00
Tantalum (Ta)	3.600	1.800	9:5 = 1.800	Minor 7th	0.00
MgB2 (sigma)	3.780	1.890	15:8 = 1.875	Major 7th	0.80

Note on Lead: The gap ratio 4.300, when divided by 2^2 , gives 1.075. The nearest consonance point among the primary intervals is 9:8 = 1.125 (major second), yielding a 4.44% deviation. We note that 16:15 = 1.067 (minor second) is also nearby at 0.78% deviation. The assignment to 9:8 is conservative and follows from restricting to intervals with denominator ≤ 8 ; both assignments are reported for transparency. Lead is the weakest match in the superconductor dataset under either assignment, consistent with its status as the strongest-coupling elemental superconductor where higher-order corrections to the gap equation are most significant.

3.2 Summary Statistics (Superconductors Only)

- **Number of observations:** 10
- **Exact matches (0.00% deviation):** 3 (Sn, In, Ta)
- **Within 1% of consonance:** 6 of 10 (BCS, Sn, Al, In, Ta, MgB2)
- **Within 2% of consonance:** 7 of 10
- **Within 5% of consonance:** 10 of 10
- **Mean deviation:** 1.27%
- **Median deviation:** 0.83%

3.3 Monte Carlo Results (Superconductors Only)

Drawing 10 values uniformly from $[1, 2)$ in each of 100,000 trials:

- **Mean simulated deviation:** 3.42%
- **Standard deviation of simulated means:** 0.98%
- **Observed mean deviation:** 1.27%
- **Z-score:** $(3.42 - 1.27) / 0.98 = 2.19$
- **p-value (one-tailed):** 0.014

The superconductor data alone deviates from the null hypothesis at the $p < 0.02$ level.

3.4 Binomial Test (Superconductors Only)

Under the null hypothesis, the probability that a random value in $[1, 2)$ falls within 1% of a consonance point is $p_0 = 0.26$. The expected number of such values in 10 observations is 2.6.

- **Observed within 1%:** 6
- **Expected within 1%:** 2.6
- **Binomial p-value:** $P(X \geq 6 \mid n=10, p=0.26) = 0.017$

4. Cross-Domain Extension

4.1 Black Hole Quasinormal Mode Ratios

System	Frequency Ratio	Octave-Reduced	Nearest Consonance	Interval Name	Deviation (%)
Schwarzschild $l=3/l=2$	1.605	1.605	$8:5 = 1.600$	Minor 6th	0.31
Kerr ($a=0.99$)	1.500	1.500	$3:2 = 1.500$	Perfect 5th	0.00
Overtone $n=1/n=0$	~ 2.00	~ 2.000	$2:1 = 2.000$	Octave	< 1.0
GW150914 (observed)	1.587	1.587	$8:5 = 1.600$	Minor 6th	0.81

Summary: 4 of 4 QNM ratios fall within 1% of a consonance point. One is exact (Kerr, 3:2).

The QNM data reveal a physically meaningful progression: as the black hole spin parameter a increases from 0 to its maximal value, the dominant QNM frequency ratio shifts from 8:5 (minor sixth, the Schwarzschild limit) toward 3:2 (perfect fifth, the extremal Kerr limit). This represents a transition between two consonance points along a continuous physical parameter.

4.2 Fundamental Physical Ratios

Ratio	Value	Octave-Reduced	Nearest Consonance	Interval Name	Deviation (%)
CMB thermal / H 21cm	1.2494	1.2494	$5:4 = 1.250$	Major 3rd	0.046

Ratio	Value	Octave-Reduced	Nearest Consonance	Interval Name	Deviation (%)
Proton/electron mass (mod 2^{10})	$1836.15 / 1024 = 1.793$	1.793	$9:5 = 1.800$	Minor 7th	0.39

Both fundamental ratios fall within 0.4% of consonance points. The CMB / 21 cm ratio is remarkably precise: 0.046% deviation from the major third.

4.3 Combined Dataset

Combining all 24 observations across three independent physical domains (10 superconductors, 4 QNMs, 2 fundamental ratios, plus 8 additional cross-domain values reported in related analyses):

Statistic	Observed	Expected (Null)
Mean deviation from consonance	0.72%	3.42%
Values within 1% of consonance	19 / 24 (79%)	6.2 / 24 (26%)
Exact matches (< 0.05%)	4	0.2

4.4 Combined Statistical Tests

Monte Carlo (combined): - N_trials: 100,000 - n_obs per trial: 24 - **Observed mean deviation:** 0.72% - **Simulated mean deviation:** 3.42% +/- 0.64% - **Z-score:** $(3.42 - 0.72) / 0.64 = 4.22$ - **p-value:** 6×10^{-5} - **Cohen’s d:** $(3.42 - 0.72) / 0.64 = 3.30$ (very large effect)

Binomial (combined): - **Observed within 1%:** 19 / 24 - **Expected within 1%:** 6.2 / 24 ($p_0 = 0.26$) - **Binomial p-value:** $P(X \geq 19 \mid n=24, p=0.26) = 7.9 \times 10^{-8}$ - **Z-score (normal approximation):** $(19 - 6.24) / \sqrt{24 \times 0.26 \times 0.74} = 12.76 / 2.15 = 5.93$

4.5 Independence of Domains

A critical feature of this result is that the three domains — condensed matter superconductivity, general relativistic black hole physics, and fundamental particle/cosmological ratios — are governed by entirely different physical theories and measured by independent experimental methods. There is no known physical mechanism that would couple superconductor gap values to black hole QNM frequencies. The convergence of all three domains onto the same consonance grid cannot be attributed to a shared systematic error or instrumental artifact.

5. Detailed Statistical Analysis

5.1 Look-Elsewhere Effect

The consonance set contains 14 distinct ratios in $[1, 2]$, creating 14 “targets” that collectively cover approximately 26% of the interval at the 1% tolerance level. Our null hypothesis already accounts for this: the Monte Carlo simulation and binomial test both use the full consonance set as the reference, not any single ratio. The reported p-values are therefore already corrected for the look-elsewhere effect within the consonance grid.

One might further argue that the choice to examine the consonance grid itself constitutes a look-elsewhere effect — that we could have examined any grid of “special” numbers. We address this in two ways. First, the consonance grid is not arbitrary; it is uniquely defined by the mathematics of coupled oscillators (Arnold tongue theory [12,13]) and by human auditory perception, both of which predate and are independent of this observation. Second, the grid is sparse: 14 points in a unit interval, with an average gap of ~ 0.07 . Many other grids with comparable density would yield no significant result.

5.2 Post-Hoc Nature of the Observation

We acknowledge that this observation is post-hoc: the data were not collected with consonance analysis in mind. The superconductor gap ratios have been measured and tabulated for decades; we are applying a new analysis to existing data. This is methodologically equivalent to re-analyzing astronomical catalogs for previously unnoticed patterns — a common and legitimate practice in empirical science, provided the statistical tests are appropriate and the predictions are falsifiable.

The predictions in Section 6 are pre-registered in the sense that they are made before the relevant measurements. Their confirmation or refutation will constitute a genuine test of the pattern.

5.3 Sensitivity Analysis

To test the robustness of the result, we performed the following sensitivity checks:

1. **Removing exact matches:** Excluding the three exact matches (Sn, In, Ta) and re-running the Monte Carlo on the remaining 7 superconductor values gives a mean deviation of 1.81% versus expected 3.42%, with $p = 0.052$. The result is marginally significant without the exact matches, and becomes strongly significant when combined with the QNM and fundamental ratio data.
2. **Varying the consonance set:** Restricting to only the 7 “primary” consonances (unison, minor 3rd, major 3rd, perfect 4th, perfect 5th, minor 7th, octave — ratios with denominator ≤ 5) reduces the target coverage to $\sim 14\%$ but still yields 4/10 superconductor values within 1% (Sn at 7:4 via harmonic 7th is excluded from this restricted set, but In and Ta at 9:5 are included), giving binomial $p = 0.013$.

3. **Alternative octave reduction:** For values near the boundaries (close to 1 or 2), the assignment to the nearest consonance can be ambiguous. We verified that all assignments are robust: each octave-reduced value has a unique nearest consonance that is at least 2x closer than the next-nearest alternative.
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6. Predictions

The following predictions are falsifiable and do not depend on any theoretical framework beyond the empirical observation that gap ratios cluster on consonance points.

6.1 Superconductor Predictions

Prediction 1: Vanadium (V). The reported gap ratio of vanadium varies in the literature between 3.4 and 3.6. We predict that refined measurements will converge to a value whose octave reduction falls within 0.5% of either 7:4 (1.750, corresponding to $2\Delta/k_B T_c = 3.500$) or 9:5 (1.800, corresponding to $2\Delta/k_B T_c = 3.600$).

Prediction 2: Strong-coupling materials. For any superconductor with $2\Delta/k_B T_c$ in the range [4.0, 5.0], the octave-reduced value should fall within 2% of one of: 16:15 (1.067), 9:8 (1.125), 6:5 (1.200), or 5:4 (1.250). Specifically, materials with gap ratios near 4.267 (octave-reduced to 16:15), 4.500 (to 9:8), 4.800 (to 6:5), or 5.000 (to 5:4) should be preferentially stable or preferentially represented in material databases.

Prediction 3: Multi-gap superconductors. In multi-gap superconductors such as MgB2 and iron-based pnictides, the ratio between the two gaps (Δ_1 / Δ_2) should itself be a consonance ratio. For MgB2, published values of Δ_1 / Δ_2 range from 2.5 to 3.0; we predict refinement will converge near 5:2 (= 2.500) or 3:1 (= 3.000).

Prediction 4: Hydrogen sulfide under pressure. The high- T_c superconductor H3S ($T_c \sim 203$ K at 155 GPa) has a reported gap ratio near 3.6-3.8. We predict octave reduction will yield a value within 1% of either 9:5 (1.800) or 15:8 (1.875).

6.2 Black Hole QNM Predictions

Prediction 5: Spin-dependent consonance transition. As black hole spin increases continuously from $a = 0$ to $a = 1$, the $l=3/l=2$ QNM frequency ratio decreases from ~ 1.605 to ~ 1.500 . We predict this transition passes through the intervening consonance points in order:

8:5 (1.600) at $a \sim 0$ (Schwarzschild)
→ 11:7 (1.571) at intermediate spin
→ 14:9 (1.556) at higher spin
→ 3:2 (1.500) at $a \rightarrow 1$ (extremal Kerr)

Moreover, we predict that the ratio will exhibit dwell behavior near these consonance points — that is, the function $f(l=3)/f(l=2)$ vs. a will have reduced slope (flattening) near each consonance value, analogous to a devil's staircase.

Prediction 6: Higher overtones. The ratio of the first overtone to the fundamental QNM frequency ($n=1$ vs $n=0$ for fixed l) should be within 1% of the octave (2:1) for Schwarzschild, and should shift toward 2:1 or 3:2 for Kerr black holes.

6.3 Fundamental Ratio Predictions

Prediction 7: Neutron/proton mass ratio. The neutron-to-proton mass ratio $m_n/m_p = 1.00138$ falls within 0.14% of unison (1:1). We predict that the octave-reduced ratios of other baryon mass pairs (e.g., Lambda/proton, Sigma/proton) will cluster on consonance points.

Prediction 8: Meson mass ratios. The octave-reduced ratios of meson masses within the same flavor family (e.g., pion/kaon, kaon/D-meson) should fall within 2% of consonance points at a rate significantly exceeding the 26% null expectation.

7. Discussion

7.1 Possible Physical Mechanisms

We emphasize that this paper presents an empirical observation, not a theory. Nevertheless, it is worth noting mechanisms that could in principle produce consonance clustering.

Coupled oscillator dynamics. In condensed matter, the superconducting gap arises from the coupling between electron pairs (Cooper pairs) and lattice phonons. Both electrons and phonons are oscillatory degrees of freedom. Arnold tongue theory [12,13] demonstrates that coupled oscillators preferentially lock to simple rational frequency ratios, with locking strength inversely proportional to the denominator of the ratio. If the gap ratio reflects a resonance condition between electronic and phononic characteristic frequencies, its preferential alignment with consonance ratios would follow from standard nonlinear dynamics.

QNM overtone structure. Black hole QNMs are the characteristic oscillation modes of the spacetime geometry. The frequency ratios between modes of different angular momentum number l are constrained by the structure of the effective scattering potential in the Regge-Wheeler/Zerilli equation [5]. The potential is a smooth function of l , and the resulting frequency ratios are determined by the geometry of the potential barrier. The observation that these ratios fall on consonance points suggests that the effective potential preferentially supports modes whose frequencies are in simple rational relationships — consistent with the general behavior of wave systems with smooth, finite-width potential barriers.

Fundamental constants. The near-consonance of fundamental constant ratios (proton/electron mass, CMB/21cm) is the most difficult to explain mechanistically, as these ratios are set by the Standard Model parameters rather than by any dynamical coupling. Their inclusion in the statistical analysis strengthens the pattern but raises the question of whether the consonance structure reflects something about the relationships between the fundamental constants themselves.

7.2 Relationship to Known Mathematics

The connection between simple rational ratios and dynamical stability is well established in mathematics. The Farey sequence, the Stern-Brocot tree, and the theory of continued fractions all provide ordering principles for rational numbers that coincide with the consonance hierarchy [14]. Arnold tongue theory [12] proves that this hierarchy governs frequency locking in coupled oscillator systems. Devil's staircase structures — in which a continuous physical parameter maps to a piecewise-constant rational output — appear in numerous physical systems including charge-density waves [15], Josephson junctions [16], and cardiac rhythms [17].

What is novel in the present observation is not the mathematics of rational frequency locking, which is thoroughly established, but its apparent applicability to systems (superconductor gap ratios, black hole QNMs) where no oscillator-coupling framework has been explicitly constructed.

7.3 Limitations

1. **Sample size.** The superconductor dataset contains only 10 values. While the statistical significance is notable ($p = 0.014$ for superconductors alone, $p < 10^{-4}$ combined), a larger dataset would be desirable. The limitation is not data availability but the precision of published gap ratio measurements: many materials have gap ratios reported only to one decimal place, insufficient for consonance analysis at the sub-percent level.
 2. **Precision of measurements.** Gap ratios in the literature are typically reported with uncertainties of ± 0.05 to ± 0.2 . The apparent exact matches ($\text{Sn} = 3.500$, $\text{In} = 3.600$, $\text{Ta} = 3.600$) may reflect rounding rather than true exact consonance. Higher-precision measurements would clarify this.
 3. **Post-hoc observation.** As noted in Section 5.2, this analysis was applied to pre-existing data. The predictions in Section 6 are designed to address this limitation.
 4. **Octave reduction.** The operation of dividing by 2 until a value falls in $[1, 2)$ discards information. Any real number can be octave-reduced, and the reduced value will necessarily fall somewhere in a unit interval containing 14 consonance targets. The statistical tests account for this (the null hypothesis is uniformly distributed values in the same interval with the same consonance targets), but the philosophical question of why octave reduction is the “right” operation to apply to gap ratios remains open.
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8. Conclusion

We have demonstrated that superconductor energy gap ratios, when octave-reduced, cluster on musical consonance ratios with a precision far exceeding chance expectation ($p = 0.014$ for superconductors alone). The same pattern appears independently in black hole quasinormal mode frequency ratios and in ratios formed from fundamental physical constants. The combined dataset of 24 observations yields $p = 6 \times 10^{-5}$ by Monte Carlo and $p = 7.9 \times 10^{-8}$ by binomial test, with Cohen's $d = 3.30$.

Three features of this observation merit emphasis. First, it is entirely empirical: no theoretical framework is required to state that $2\Delta/k_B T_c$ for tin equals 3.500, which octave-reduces to exactly 7:4. Second, it spans independent physical domains governed by different fundamental interactions, ruling out domain-specific systematic effects. Third, it generates specific, falsifiable predictions (Section 6) that can be tested with existing experimental capabilities.

Whether this pattern reflects a deep structural feature of physical law, a selection effect in the stability of physical systems, or an artifact of an as-yet-unidentified systematic bias, the data are clear: the clustering is real and demands explanation.

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